

Swinburne University of Technology Hawthorn Campus
Dept. of Computer Science and Software Engineering

COS10022 Data Science Principles

Assignment 1 - *Semester 1, 2026*

Assessment Title: Predictive Model Creation and Evaluation

Assessment Weighting: 20%

Due Date: Friday, 27th March 2026 at 11.59 pm (AEDT)

Assessable Item:

- One (1) piece of a written report, no more than 10 pages.
- One (1) zip file containing your KNIME workflow, the input file, the output file or any intermediate files produced in your workflow execution process.
- One (1) video introducing the chain of thought in which you used to complete the task, the challenge and the solutions you come up with for this assignment, and the takeaway points of what you learned in this assignment.
- The submitted report must pass the Turnitin check on Canvas with no more than 25% similarity except for the parts from the template or the short answers.

The submitted report should answer all questions listed in the assignment task section in sequence. Submitting the zip file containing the explanation video, the input/output files and your fully functional KNIME workflow is essential for your submission to be marked. If the submitted zip file cannot execute properly or the execution result differs from what you have in the report, you will not get the mark, even if you submitted the report.

Purpose of Assignment

This assignment aims to evaluate students' achievement of the following unit learning outcomes:

1. **Appreciate (and explain) the key concepts, techniques, and tools for handling the data and creating prediction models.**
2. **Experiencing model selection and implementation for carrying out a data science project.**
3. **Integrating Generative Artificial Intelligence (genAI) in the task-solving process to aid professional skills.**

This is an individual assignment. Refer to the Unit Outline for the late submission penalty policy. You can ignore the high similarity that appears on the cover page, the template wording, and the short answers. You must make sure your submitted report has a similarity lower than 25% in total and less than 5% from a single source. Otherwise, your report will not be marked.

Key Lessons:

You are asked to use the specified dataset and then build models in KNIME analytic platform to maximise your test accuracy and explain your design concept.

Introduction

This dataset contains detailed records of simulated road accident data, focusing on factors influencing survival outcomes. The dataset includes demographic, behavioural, and situational attributes, providing valuable insights into how various factors impact the survival probability during road accidents. You are expected to follow the instructions for building your predictive model and answer questions.

Assignment Goal

This assignment aims at building experiences for students to train a usable predictive model, and explain the outputs. A small part of discovery and research component is included in the assignment for expending the skill set of the students. GenAI usage is also part of the skill training for assisting with daily work in the data science field.

Assignment Task

The dataset has been cleaned and organised with no missing data. Your tasks are to create predictive models and to prepare the data according to the instructions to answer the questions listed below. The source files are specified in the tasks. The report should be prepared in a neat, easy-to-read style, with clear information that answers the questions. A table of Contents is not required.

The data source contains many details of the record. We aim to build predictive models that indicate whether an individual survived an accident by feeding all attributes into the model. Please follow the steps below to prepare your data after loading it into KNIME. When preparing the zip file, please place all input files in the same directory as your KNIME workflow. This will make the marking easier for the tutors. You can keep growing the workflow across multiple steps or start an independent one, depending on your need to complete the tasks:

1. Use the "accident_2025_train2_73.csv" and the "accident_2025_test2_73.txt" to be the training and the test set, respectively. Build a Logistic Regression Model to predict an individual's survival.
   Things to be included in the report: Explain how you came up with the workflow, what the detailed configurations are, and the reason you made the configuration. Describe what you see in your prediction results, what difficulties you encountered when loading the files, and how you solved the issues.
2. Use all CSV files in the "src" folder as the input. (Please note that you cannot modify the file structure of the "src" folder. You must use nodes in KNIME to process everything.) Read and learn how to use the "Rule-based Row Splitter" in KNIME to put tuples with the same value of the "Survived" label together. Use the **workflow coach** at one of the rule-based row splitter node outputs to select the appropriate node to partition the data into two groups. One of the groups should contain exactly 350 tuples, and the other should contain the remaining tuples. Read and learn how to use the "Concatenate" node to concatenate both datasets with exactly 350 tuples together to form an Independent and Identically Distributed (IID) dataset with 700 tuples to be the training set. The remaining tuples should be concatenated together as the test set. Both datasets should be shuffled before being fed into the predictive models. Create a logistic regression, a Naïve Bayes, and a random forest model with identical training and test sets to compare their performance.
   Things to be included in the report: Describe how you managed to take in all CSV files as the input, what rules are set in the rule-based row splitter, what prompt was used to find the appropriate node for partitioning the data, and reveal the detailed configuration of nodes used to prepare the data. Report any issues encountered during training the prediction models, along with your solutions. Compare the results obtained by these models and explain your insights and suggestions.

Submission Requirement

To fulfil the requirement of this assignment, the following items should be prepared in MS Word or PDF format. Most of the texts in the report should be typed, not converted into images. **The report should contain the following information on the top of the first page: your name, student ID, unit code, class (CL1/0x), and tutor's name.**

Submitting the zip file containing the input/output files and your fully functional KNIME

workflow is essential for your submission to be marked. If the submitted zip file cannot execute properly or the execution result differs from what you have in the report, you will not get the mark.

Failure to adhere to the submission requirements will immediately result in no mark for this assignment.

Rubric

Criterion	Excellent (HD)	Good (D)	Satisfactory (C)	Limited (P)	Poor (F)	Weight
Understanding of Workflow Design	Workflow is logically structured, clearly explained step-by-step, and demonstrates strong understanding of data science workflow design principles.	Workflow is mostly clear with minor gaps in explanation.	Workflow is described but lacks depth or clarity in reasoning.	Workflow partially explained; structure somewhat unclear.	Workflow is poorly described or missing.	5%
Detailed Node Configuration	All nodes are fully documented with precise configuration details and justification for each parameter.	Most nodes documented with reasonable explanation.	Basic node configuration described but limited explanation.	Minimal description of node configuration.	Node configurations missing or incorrect.	15%
Data Loading and Pre-processing	Demonstrates correct loading of datasets, clear explanation of issues encountered (e.g., file format, delimiter, type mismatch) and effective solutions.	Minor issues discussed with reasonable solutions.	Basic explanation of loading process.	Limited discussion of issues or solutions.	No explanation or incorrect handling of data loading.	10%
Rule-based Row Splitter Implementation	Rules clearly defined, logically justified, and correctly	Rules implemented correctly with some explanation.	Rules used but explanation limited.	Incorrect or unclear rule definition.	Rule-based splitting not implemented.	5%

Criterion	Excellent (HD)	Good (D)	Satisfactory (C)	Limited (P)	Poor (F)	Weight
	implemented to group tuples with the same **Survived** value.					
Dataset Partitioning Process	Correct use of KNIME nodes to produce exactly **350 tuples per subset** and correct concatenation into **IID 700 training set** and test set; process clearly explained.	Minor inaccuracies but overall correct implementation.	Partitioning achieved but explanation lacks clarity.	Partitioning partially incorrect or poorly explained.	Partitioning not correctly implemented.	10%
Use of KNIME Workflow Coach & Node Selection	Prompt used with workflow coach clearly described; node selection well justified.	Prompt mentioned but explanation limited.	Node selection described but workflow coach usage unclear.	Minimal explanation of node selection.	No evidence of workflow coach usage.	5%
Model Construction	Logistic Regression, Naïve Bayes, and Random Forest correctly built with identical training/test datasets and appropriate configuration.	All models implemented with minor configuration issues.	Models built but configuration explanation limited.	Only some models correctly implemented.	Models missing or incorrectly implemented.	5%
Model Evaluation and Results	Results clearly presented (accuracy, confusion matrix, etc.) with correct interpretation.	Results presented with some analysis.	Results reported but interpretation shallow.	Results incomplete or unclear.	No evaluation results.	10%
Comparison and Insights	Strong analytical comparison of models with meaningful insights into performance differences.	Comparison present with reasonable explanation.	Basic comparison provided.	Minimal comparison.	No comparison.	10%

Criterion	Excellent (HD)	Good (D)	Satisfactory (C)	Limited (P)	Poor (F)	Weight
Problem Solving & Troubleshooting	Multiple issues identified with clear explanation of debugging process and solutions.	Some issues identified and addressed.	Basic discussion of challenges.	Limited troubleshooting discussion.	No troubleshooting described.	5%
Report Quality & Presentation	Report is professionally structured with clear sections, logical flow, and concise academic writing. Figures, tables, and KNIME screenshots are well-labelled and effectively support explanations. No or negligible language errors.	Report is well organized with clear headings and generally good writing. Figures and screenshots are included but may lack some labeling or detailed explanation. Minor language errors present.	Report structure is adequate but sometimes unclear. Some screenshots or figures are provided but not always well integrated into the discussion. Several language or formatting issues.	Report structure is weak or inconsistent. Figures or screenshots are limited or poorly explained. Writing clarity is affected by frequent language or formatting problems.	Report is poorly organized or incomplete. Little or no supporting figures/screenshots. Writing is unclear and difficult to follow.	10%
Exploration & Discovery in Self-Learning	Demonstrates strong independent learning by exploring both technical tools and underlying data science concepts. Clearly explains discoveries related to workflow design, model behaviour, and theoretical concepts. Shows initiative in experimenting with multiple approaches	Shows clear effort in self-learning beyond basic instructions. Demonstrates understanding of relevant data science concepts and explains how these ideas influenced workflow design or model training. Some reflection on the learning process.	Demonstrates some independent exploration of tools or concepts. Mentions relevant data science ideas but explanations are brief or partially developed. Limited reflection on learning process.	Minimal evidence of independent exploration. Focuses mainly on following instructions without demonstrating deeper understanding of data science principles behind the workflow.	No evidence of self-learning or conceptual exploration. Work shows little understanding of either the tools used or the underlying data science concepts.	10%

Criterion	Excellent (HD)	Good (D)	Satisfactory (C)	Limited (P)	Poor (F)	Weight
	and reflects critically on what was learned.					

---------- End of Assignment ----------